WORKED SOLUTION

STEP III 1998, Question 4

Pure • Polar coordinates

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s3-q4

The key idea

Polar curves are easiest to understand by converting to Cartesian form when the equation is simple, and by tracking the sign of r when it isn't. The whole question hinges on the convention that a point with $r < 0$ is plotted at distance $|r|$ in the *opposite* direction, $\theta + \pi$. For $r = \cos k\theta$ that convention decides whether the negative loops land on fresh ground (giving new petals) or on top of petals already drawn.

Solution

The circle $r = \cos\theta$

Multiply both sides by r and use $x = r\cos\theta$, $y = r\sin\theta$, $r^2 = x^2 + y^2$:

$$\begin{aligned} r^2 = r\cos\theta \quad &\Longrightarrow \quad x^2 + y^2 = x \\ &\Longleftrightarrow \quad \left(x - \tfrac{1}{2}\right)^2 + y^2 = \tfrac{1}{4}. \end{aligned}$$

So every point of the curve lies on the circle with centre $(\frac{1}{2}, 0)$ and radius $\frac{1}{2}$.

Conversely, take any point (x, y) of this circle other than the origin. Since $x = x^2 + y^2 > 0$, its polar angle satisfies $-\frac{1}{2}\pi < \theta < \frac{1}{2}\pi$, and dividing $r^2 = x = r\cos\theta$ by $r > 0$ gives $r = \cos\theta$. The origin itself is reached at $\theta = \pm\frac{1}{2}\pi$. Hence the curve is exactly this circle, traced once as θ runs from $-\frac{1}{2}\pi$ to $\frac{1}{2}\pi$, and on this range $r = \cos\theta \geq 0$ throughout, so no sign convention is needed.

The curve $r = \cos 2\theta$

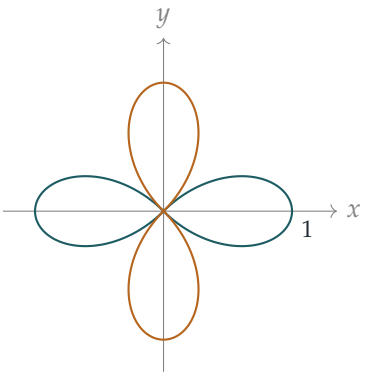
Now r changes sign, and we plot (r, θ) with $r < 0$ at distance $|r|$ in direction $\theta + \pi$.

- For $-\frac{1}{4}\pi < \theta < \frac{1}{4}\pi$, $r > 0$, rising from 0 to 1 at $\theta = 0$ and back to 0: a petal along the positive x-axis.
- For $\frac{1}{4}\pi < \theta < \frac{3}{4}\pi$, $r < 0$, so these points are drawn in directions $\frac{5}{4}\pi$ to $\frac{7}{4}\pi$: a petal

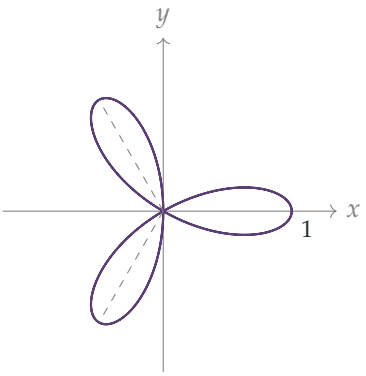
along the negative y-axis.

- Similarly $\frac{3}{4}\pi < \theta < \frac{5}{4}\pi$ $(r > 0)$ gives a petal along the negative x-axis, and $\frac{5}{4}\pi < \theta < \frac{7}{4}\pi$ $(r < 0)$ gives one along the positive y-axis.

The result is a four-leaved rose, each petal of length 1, symmetric about both axes and touching the others only at the origin. In the sketch, teal petals come from $r > 0$ and amber petals from $r < 0$.



$r = \cos 2\theta$



$r = \cos 3\theta$

The curves $r = \cos 2n\theta$

Since $\cos(-2n\theta) = \cos 2n\theta$, n and $-n$ give the same curve, so take $n \geq 1$ (we come back to $n = 0$ below).

> **Ask yourself** When $r < 0$, do the loops land on top of the ones already drawn, or somewhere new?

We have $r > 0$ on intervals of length $\pi/(2n)$ centred at $\theta = k\pi/n = 2k \cdot \frac{\pi}{2n}$, for $k = 0, 1, \ldots, 2n-1$. Each gives a petal pointing in the direction $2k \cdot \frac{\pi}{2n}$, an *even* multiple of $\frac{\pi}{2n}$.

We have $r < 0$ on intervals centred at $\theta = (2k+1)\frac{\pi}{2n}$, and these points are plotted in direction $\theta + \pi$, so the petal points along

$$(2k+1)\frac{\pi}{2n} + \pi = (2k+1+2n)\frac{\pi}{2n},$$

an *odd* multiple of $\frac{\pi}{2n}$. So the negative loops land somewhere new, in the gaps between the positive petals.

Hence $r = \cos 2n\theta$ is a rose with $4n$ congruent petals, each of length 1, pointing along the directions $m\pi/(2n)$ for $m = 0, 1, \ldots, 4n-1$, each petal occupying an angle $\pi/(2n)$, and neighbouring petals meeting only at the origin. (For $n = 1$ this is the four-leaved rose above.) The petals have total angle $4n \cdot \frac{\pi}{2n} = 2\pi$, so they fill the full turn with no overlap.

For the degenerate case $n = 0$ the curve is $r = 1$, the unit circle.

The area is independent of n

Because the petals don't overlap, each part of the enclosed region is swept exactly once as θ goes from 0 to 2π, and the area formula $\frac{1}{2}\int r^2\,\mathrm{d}\theta$ (which only involves r^2, so the sign of r doesn't matter) gives, for $n \geq 1$,

$$\begin{aligned} A &= \frac{1}{2}\int_0^{2\pi} \cos^2 2n\theta\,\mathrm{d}\theta = \frac{1}{4}\int_0^{2\pi} (1+\cos 4n\theta)\,\mathrm{d}\theta \\ &= \frac{1}{4}\left[\theta + \frac{\sin 4n\theta}{4n}\right]_0^{2\pi} = \frac{\pi}{2}, \end{aligned}$$

since $\sin 8n\pi = 0$. This doesn't depend on n, as required: adding more petals makes each petal thinner in exact compensation. (The excluded case $n = 0$, the unit circle, has area π; the formula's $\sin 4n\theta/(4n)$ term is exactly where $n \neq 0$ is used.)

The curve $r = \cos 3\theta$

Ask yourself Why does this curve have three petals, not six?

Now $r > 0$ on intervals centred at $\theta = 0, \frac{2}{3}\pi, \frac{4}{3}\pi$, giving petals in those directions. The negative loops are centred at $\theta = \frac{1}{3}\pi, \pi, \frac{5}{3}\pi$, and are plotted in directions $\frac{4}{3}\pi$, 2π and $\frac{8}{3}\pi \equiv \frac{2}{3}\pi$: exactly the directions of the positive petals. Indeed, replacing θ by $\theta + \pi$ gives

$$r = \cos(3\theta + 3\pi) = -\cos 3\theta$$

in the opposite direction, which is the same point. So $r = \cos 3\theta$ is a three-leaved rose, petals of length 1 along $\theta = 0, \frac{2}{3}\pi, \frac{4}{3}\pi$, each occupying an angle $\frac{1}{3}\pi$, and the whole curve is traced *twice* as θ runs from 0 to 2π (once for $0 \leq \theta \leq \pi$). This is the sketch on the right above.

The same argument shows that for any odd k, $r = \cos k\theta$ is a k-leaved rose traced twice, while for even k it is a $2k$-leaved rose traced once.

Checks

- Put $\theta = 0$ in $r = \cos\theta$: the point $(1, 0)$ is on $(x - \frac{1}{2})^2 + y^2 = \frac{1}{4}$, and so is the origin (at $\theta = \pm\frac{1}{2}\pi$).
- The circle $r = \cos\theta$ has area $\pi(\frac{1}{2})^2 = \frac{1}{4}\pi$, and $\frac{1}{2}\int_{-\pi/2}^{\pi/2} \cos^2\theta\,\mathrm{d}\theta = \frac{1}{4}\pi$ agrees. Similarly $\frac{1}{2}\int_0^{\pi} \cos^2 3\theta\,\mathrm{d}\theta = \frac{1}{4}\pi$ for the three-leaved rose. Using 0 to 2π there would double count and give $\frac{1}{2}\pi$.
- A numerical check: plotting $r = \cos 2n\theta$ for $n = 1, \ldots, 5$ finds $4n$ petal tips, and a Monte Carlo estimate of the enclosed area gave $1.57 \approx \pi/2 = 1.5708$ every time.

Common slips

- Ignoring negative r and drawing only two petals for $r = \cos 2\theta$.
- Saying that $r = \cos 3\theta$ has six petals because it has six loops of r: the negative loops land on top of the positive ones.
- Using $\frac{1}{2}\int_0^{2\pi} r^2\,\mathrm{d}\theta$ for a curve traced twice (such as $r = \cos 3\theta$), which doubles the area.

Beyond STEP

These curves are the *rhodonea* or rose curves. Writing a point as the complex number $z = r\,\mathrm{e}^{\mathrm{i}\theta}$, the rose $r = \cos k\theta$ is $z = \frac{1}{2}\left(\mathrm{e}^{\mathrm{i}(k+1)\theta} + \mathrm{e}^{-\mathrm{i}(k-1)\theta}\right)$, the sum of two circular motions in opposite directions. This viewpoint links roses to epicycles and to Fourier series, and the area formula $\frac{1}{2}\int r^2\,\mathrm{d}\theta$ is a special case of Green's theorem in vector calculus. Needham's *Visual Complex Analysis* develops this geometric way of seeing complex numbers beautifully.

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